

Number sense, generations, abbreviations and security recap

One college period · 40–45 minutes · Paper I mixed theory

What you will learn

- Recall computer generations
- Expand the abbreviations that appear every year
- Restate virus / antivirus and basic security
- Keep a one-page formula sheet for Paper I

Generation	Technology	Example / note
1st (1940s–50s)	Vacuum tubes	ENIAC, huge, hot
2nd	Transistors	Smaller, more reliable
3rd	Integrated circuits (ICs)	Board MCQ favourite
4th	Microprocessor	IBM PC, personal computers
5th	AI, ULSI, natural language	Modern research / assistants

Full forms to memorise

Abbrev.	Full form
CPU	Central Processing Unit
ALU / CU	Arithmetic and Logic Unit / Control Unit
RAM / ROM / BIOS	Random Access Memory / Read Only Memory / Basic Input Output System
USB / HDMI / VGA	Universal Serial Bus / High-Definition Multimedia Interface / Video Graphics Array
OCR / OMR / MICR / CRT	Optical Character / Mark / Magnetic Ink Character Recognition / Cathode Ray Tube
GUI / CLI / OS	Graphical User Interface / Command Line Interface / Operating System
LAN / MAN / WAN / NIC	Local / Metropolitan / Wide Area Network / Network Interface Card
OSI / ISO / IEEE / SMTP / FTP / HTTP / DNS / URL / IP / TCP	Open Systems Interconnection / International Organization for Standardization / Institute of Electrical and Electronics Engineers / Simple Mail Transfer Protocol / File Transfer Protocol / HyperText Transfer Protocol / Domain Name System / Uniform Resource Locator / Internet Protocol / Transmission Control Protocol
SIMM / SRAM / DRAM / EEPROM	Single Inline Memory Module / Static / Dynamic RAM / Electrically Erasable PROM
AI / ICT / IT	Artificial Intelligence / Information and Communication Technology / Information Technology

REMEMBER IT

Every night before the paper, write this table from memory. Section B often says “write full form of any three”.

CHECK YOURSELF

1. 3rd generation technology?
2. BIOS full form?

Answers: 1. Integrated circuits · 2. Basic Input Output System

MCQs

1. OCR stands for:
 - a. Optical Character Recognition
 - b. Original Code Reader
 - c. Output Card Reader
 - d. Open Computer RAM
2. GUI stands for:
 - a. General User Internet
 - b. Graphical User Interface
 - c. Global Unique ID
 - d. General Utility Input

Key: 1=a, 2=b

Short questions (3 marks each)

1. Write full forms of OCR, GUI, BIOS, SMTP, USB.
2. Write a short note on computer generations.

Model short answers

1. OCR Optical Character Recognition; GUI Graphical User Interface; BIOS Basic Input Output System; SMTP Simple Mail Transfer Protocol; USB Universal Serial Bus.
2. Five generations from vacuum tubes to microprocessors and AI, each smaller, faster and cheaper.